

1. Introduction

I chose the Kickstarter dataset because it suits the idea of my project in predicting product success before they hit the market.

2. Dataset Description

The dataset contains 378,661 rows and 15 columns, including features like category, goal, and pledged amount. The target column is 'state', which represents whether the project succeeded or failed.

3. Models Used

I trained three models: Logistic Regression Accuracy = 0.5759, Random Forest Accuracy = 1.0000, and KNN Accuracy = 0.4940. Random Forest was the best model among them.

4. Neural Network

My neural network had two Dense layers with 128 and 64 neurons, and a Dropout layer of 0.3. It was trained for 10 epochs using sparse categorical crossentropy loss, and reached a validation accuracy of 52.56%.

5. Results & Comparison

The Random Forest model achieved a perfect score of 100% because it easily found the target variable left in the data. The Neural Network performed much worse (52.56%) because it struggled with large, unscaled financial numbers.

6. What I Learned

I learned that complex neural networks are not always better than simple models if the data is not scaled. I also learned that accidentally leaving the target column in the features leads to data leakage and fake perfect scores.